

Theoretical Bounds and Mechanisms for Neural Scaling Laws: A Critical Survey

Research synthesis

August 15, 2026

Abstract

Neural scaling laws are empirical regularities: over substantial but finite ranges, held-out loss often decreases approximately as a power of model size, training data, or compute. This survey separates exact results in stylized models from heuristic transfers to modern neural networks. Rigorous examples include Gaussian rate–distortion asymptotics, minimax nonparametric rates, kernel and linear-regression learning curves under specified spectral assumptions, and worst-case approximation lower bounds. Other influential accounts—manifold tiling, discrete “quanta,” and feature superposition—are mechanisms or hypotheses rather than universal theorems. A 2026 theory of Cagnetta, Raventós, Ganguli, and Wyart connects data-limited language-model exponents to measured conditional-entropy and token-correlation decay under an explicit horizon-limited assumption, and is currently the closest thing to a parameter-free prediction of a real exponent. The evidence nonetheless does not support a single information-theoretic law that fixes real-world exponents. Parameter count is not automatically a coding rate, a fitted loss floor is not automatically the entropy rate of language, and kernel spectra do not determine learning curves without target alignment, noise, and regularization. The most defensible conclusion is conditional: power laws arise naturally from scale-free spectra, smoothness classes, long-range dependencies, or heavy-tailed task frequencies, but which mechanism controls a trained model is an empirical and model-specific question. Throughout, we state for each framework what is proved, what is assumed, and what would have to be measured to make it predictive.

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1 Scope, terminology, and standards of evidence

Let N denote a model-size measure, D the number of training examples or tokens, and C a training-compute estimate. A common empirical fit is

$$L(N, D) = E + AN^{-\alpha_N} + BD^{-\alpha_D}. \quad (1)$$

Equation (1) is an *ansatz*, not a general law. Its fitted constants depend on the data distribution, tokenizer, context length, model family, optimization protocol, compute accounting, and fitting window. Hestness et al. [13] and Rosenfeld et al. [31] documented predictable power-law error curves across several domains before the language-model literature consolidated around Equation (1); Kaplan et al. [20] and Hoffmann et al. [15] established its practical value for compute allocation, and later replication and ablation work showed that fitting and training choices matter [3, 28].

Notation. Symbols are reused across the literature with incompatible meanings, so this survey fixes the following conventions and flags departures explicitly. Throughout, N is model size, D is data size, C is compute, and n is a generic sample size or block length inside a stylized model. Exponents attached to a resource carry a subscript (α_N, α_D); bare Greek letters are local to the subsection that introduces them. Where a cited paper uses P for tokens, or M and N for parameters and samples respectively, the statement is translated into the conventions above and the original notation is noted.

This survey uses the following evidence hierarchy.

1. **Rigorous conditional result:** a theorem for a fully specified source, function class, loss, algorithm, or asymptotic regime.
2. **Controlled-model explanation:** an analytic or numerical result in a teacher–student, kernel, random-feature, or discrete toy model.
3. **Mechanistic hypothesis:** a plausible transfer from a controlled model to trained finite neural networks.
4. **Empirical fit:** a description of observed runs over a stated range, with no claim of asymptotic universality.

Central qualification

No cited result proves that real language-model scaling is universally caused by rate–distortion, a data-manifold dimension, a neural-tangent-kernel spectrum [17], or a Zipf-like distribution of skills [27]. Each framework becomes predictive only after assumptions are supplied and relevant exponents are measured or otherwise specified.

2 What the empirical laws establish

Hoffmann et al. fit Equation (1)—writing α and β for the exponents this survey calls α_N and α_D —to their language-model experiments, reporting the rounded values $E = 1.69$, $A = 406.4$, $B = 410.7$, $\alpha_N = 0.34$, and $\alpha_D = 0.28$ [15]. Minimizing the fitted expression at fixed $C \simeq 6ND$ gives the stationarity condition $\alpha_N AN^{-\alpha_N} = \alpha_D BD^{-\alpha_D}$, hence

$$N_{\text{opt}} \propto C^{\alpha_D/(\alpha_N+\alpha_D)}, \quad D_{\text{opt}} \propto C^{\alpha_N/(\alpha_N+\alpha_D)}, \quad (2)$$

which for the rounded exponents above is $C^{0.45}$ and $C^{0.55}$, against the $C^{0.46}$ and $C^{0.54}$ that Hoffmann et al. report from their unrounded fit: nearly equal power-law growth of compute-optimal model size and token count. This is the source of the Chinchilla prescription, and Equation (2) makes clear what it depends on. It is not a universal constant-ratio theorem: the fit is tied to a particular data mixture and training recipe, and the exponents enter only through the ratio α_N/α_D , which the fit determines far less precisely than the reported digits suggest. Besiroglu et al. reconstructed the underlying data and found that the third estimation procedure of Hoffmann et al. does not fit it well, is inconsistent with their own first two procedures, and reports confidence intervals too narrow to be supported by the number of runs performed [3]. Porian et al. showed that last-layer compute accounting, warmup duration, and scale-dependent optimizer tuning explain much of the earlier discrepancy between Kaplan-style and Chinchilla-style allocations [28].

Two distinctions are essential.

- **Loss versus capability metrics.** Smooth average log loss can coexist with sharp changes in thresholded accuracy. Reported “emergent” capabilities [41] need not imply a discontinuity in the underlying predictive distribution: Schaeffer et al. show that discontinuous and nonlinear metrics such as exact-match accuracy can manufacture apparent jumps from smooth per-token improvements, and that continuous metrics on the same runs often scale smoothly [32]. This does not settle whether any particular capability is discontinuous; it shows that the measurement model must be stated before the question is meaningful.
- **Interpolation versus extrapolation.** A straight line in a log-log plot supports a power law over the observed range. It does not show that the same exponent persists to arbitrarily large scale or all the way to a loss floor; curves that look like clean power laws over a decade can break, plateau, or steepen outside the fitted window [5].

3 Information theory: exact statements and limits of the analogy

3.1 Cross-entropy and the Bayes floor

For a fixed evaluation distribution P^* , context variable Z_t (the conditioning information available at step t), next token X_{t+1} , and predictive distribution Q , expected log loss decomposes exactly as

$$\begin{aligned} \mathbb{E}_{P^*}[-\log Q(X_{t+1} | Z_t)] &= H_{P^*}(X_{t+1} | Z_t) \\ &+ \mathbb{E}_{Z_t} D_{\text{KL}}(P^*(\cdot | Z_t) \| Q(\cdot | Z_t)). \end{aligned} \quad (3)$$

Because the divergence term is nonnegative and vanishes only at $Q = P^*$, the conditional entropy is exactly the Bayes risk for log loss with that conditioning information. For a stationary source on a finite alphabet, $H(X_{t+1} | X_1^t)$ is nonincreasing in t and therefore converges to the entropy rate $h = \lim_t H(X_{t+1} | X_1^t)$ [8, 33]. Real corpora are not stationary, and h is then a property of an assumed idealization rather than a measurable constant.

The fitted constant E in Equation (1) need not equal h , and each of the following gaps is one-directional. A finite context window of length T has floor $H(X_{t+1} | X_{t-T+1}^t) \geq h$; restricted architecture, optimization error, regularization, or distribution shift can raise the asymptote further; and fitting E jointly with A, B, α_N, α_D over a finite window estimates the intercept of an extrapolated curve, not a limit that was observed. Moreover, entropy per token depends on the tokenizer, so E is not even dimensionally comparable across tokenizations without rescaling by tokens per character. Shannon’s character-prediction experiment gave bounds and estimates on a character scale [34]; modern token losses cannot be compared to those numbers without

converting both measurements on the same corpus. Compression and extrapolation estimates also depend on corpus and modeling assumptions [39].

3.2 Channel reliability is not a neural scaling law

For a discrete memoryless channel used below capacity, coding theorems yield exponential-in-blocklength behavior for achievable or optimal decoding error in appropriate rate regimes. That statement concerns communication over repeated noisy channel uses. Neural scaling curves usually concern expected prediction loss as a function of resources. The two quantities have different operational meanings, and no change of axes turns one theorem into the other.

Rate–distortion theory provides a closer mathematical analogy because it studies the smallest expected distortion attainable under a representation-rate constraint. The following example is exact.

Theorem 3.1 (Gaussian reverse water-filling example). Let X be a zero-mean Gaussian random element of a separable Hilbert space whose covariance eigenvalues are

$$\lambda_k = ck^{-(1+\rho)}, \quad c > 0, \rho > 0, \quad (4)$$

so that $\sum_k \lambda_k < \infty$. Under squared-error distortion the distortion–rate function is given parametrically by reverse water-filling,

$$D(\theta) = \sum_{k \geq 1} \min(\theta, \lambda_k), \quad R(\theta) = \frac{1}{2} \sum_{k \geq 1} \log_2^+ \frac{\lambda_k}{\theta}, \quad (5)$$

with R measured in bits. As $R \rightarrow \infty$, $D(R) \asymp R^{-\rho}$.

Proof. Only the modes with $\lambda_k > \theta$ are active, so their number is $K = K(\theta) = \lfloor (c/\theta)^{1/(1+\rho)} \rfloor \asymp (c/\theta)^{1/(1+\rho)}$; equivalently $\theta \asymp cK^{-(1+\rho)}$ and $\lambda_k/\theta \asymp (K/k)^{1+\rho}$ for $k \leq K$. Hence

$$R(\theta) \asymp \frac{1+\rho}{2} \sum_{k=1}^K \log_2 \frac{K}{k} = \frac{1+\rho}{2} \log_2 \frac{K^K}{K!} = \frac{(1+\rho) \log_2 e}{2} K + O(\log K) \quad (6)$$

by Stirling’s formula, so $R(\theta) \asymp K(\theta)$: each additional active mode buys a bounded number of bits. For the distortion, the active modes contribute $K\theta \asymp cK^{-\rho}$ and the inactive tail contributes $\sum_{k>K} ck^{-(1+\rho)} \asymp (c/\rho)K^{-\rho}$, so

$$D(\theta) = K(\theta)\theta + \sum_{k>K(\theta)} ck^{-(1+\rho)} \asymp K(\theta)^{-\rho} \asymp \theta^{\rho/(1+\rho)}. \quad (7)$$

Eliminating θ through $R \asymp K$ yields $D(R) \asymp R^{-\rho}$. $\square$

Two features of Theorem 3.1 are worth isolating, because they recur in every mechanism surveyed below. First, the power law comes entirely from the polynomial tail of the spectrum: a spectrum with a geometric tail would give exponential decay instead, as the finite-dimensional case below shows. Second, the exponent is the spectral tail exponent minus one, a property of the source alone—not of the coder, the loss scale, or the dimension of any ambient space.

For a fixed d -dimensional nonsingular Gaussian source, all d modes are active at sufficiently high rate and $D(R)$ is proportional to $2^{-2R/d}$ up to a covariance-dependent constant [2, 8].

Remark 3.1 (What this does not prove). The example shows that a heavy spectral tail *can* produce polynomial distortion–rate behavior. It does not identify neural parameter count N with a communication rate R . Such an identification requires an encoder/decoder, a bit-precision or description-length budget, a distortion matching the training loss, and an argument that training approaches the corresponding optimum. Power laws also arise from other mechanisms discussed below, so they are not a unique fingerprint of infinite-dimensional Gaussian sources.

3.3 Metric entropy and minimax estimation

For a function class $\mathcal{P}$ equipped with a metric ϱ , write

$$\mathcal{H}(\varepsilon, \mathcal{P}, \varrho) = \log \mathcal{N}(\varepsilon, \mathcal{P}, \varrho) \quad (8)$$

for its covering entropy, where $\mathcal{N}$ is the ε -covering number. Yang and Barron relate minimax rates to global and local metric entropy through information-theoretic upper bounds and testing lower bounds [42]; see also [40] for the accompanying reduction and two-point techniques. The mnemonic balance

$$\mathcal{H}(\varepsilon_n) \asymp n\varepsilon_n^2 \quad (9)$$

is valid in many regular nonparametric problems, but it is not a theorem requiring only a covering-number asymptotic: the loss, KL control, and local packing/testing conditions matter, and the balance can fail for classes whose local and global entropies have different orders.

For standard Sobolev classes of smoothness s on a d -dimensional compact domain, $\mathcal{H}(\varepsilon) \asymp \varepsilon^{-d/s}$ in the relevant norm, and Equation (9) then yields the familiar squared-risk rate

$$\varepsilon_n^2 \asymp n^{-2s/(2s+d)}. \quad (10)$$

Here n is the number of i.i.d. observations, which corresponds to D rather than N in the notation of Equation (1). This is a minimax result for a specified function class and observation model. Applying it to a language model requires justifying the effective class, metric, dependence structure, and estimator, and the resulting exponent depends on the ratio s/d , neither factor of which has an agreed operational definition for next-token prediction.

3.4 Information-theoretic model/data allocation

Jeon and Van Roy develop a Bayesian information-theoretic decomposition for prediction with a compressed or misspecified latent model [18]. In schematic form, their bound splits the average excess log loss over T observations into

$$\mathcal{E}_T \lesssim \underbrace{\frac{I(F; \tilde{F})}{T}}_{\text{estimation/information term}} + \underbrace{\mathbb{E} D_{\text{KL}}(P_F(\cdot | X) \| P_{\tilde{F}}(\cdot | X))}_{\text{misspecification term}}. \quad (11)$$

Here F is the data-generating function, $\tilde{F}$ its compressed or misspecified surrogate, T the number of observations, and $\mathcal{E}_T$ the per-observation excess loss; the first term is the cost of identifying $\tilde{F}$ from data and the second the cost of $\tilde{F}$ not being F , so the two move in opposite directions as the surrogate is made richer. For their infinite-width two-layer-network data-generating process, optimizing a derived upper bound under a product compute proxy yields an optimal relation between data size and model size that is linear up to logarithmic factors. This is a rigorous result for their Bayesian model and an ideal (Bayes-optimal) predictor; it is not a proof of Chinchilla scaling for SGD-trained transformers, and the agreement with the empirical Chinchilla ratio should be read as consistency rather than derivation. It is also worth noting what carries the conclusion: the compute-allocation result follows from their model-specific analysis, not from the generic mutual-information rate bounds that are often quoted alongside it.

4 Language statistics and data-limited scaling

4.1 Hilbert-type entropy growth

A precise version of a Hilbert-type ansatz writes block entropy as

$$H(X_1^n) = hn + cn^\delta + o(n^\delta), \quad 0 < \delta < 1. \quad (12)$$

Taking first differences gives

$$H(X_{n+1} | X_1^n) - h \asymp n^{-(1-\delta)} \quad (13)$$

when the asymptotic expansion is sufficiently regular—that is, when the expansion may be differenced term by term, which is an assumption about the source and not a consequence of Equation (12) alone. Adjacent-block mutual information, which for a stationary source equals $I(X_1^n; X_{n+1}^{2n}) = 2H(X_1^n) - H(X_1^{2n})$, also inherits an n^δ contribution under Equation (12). These relationships follow from the block-entropy ansatz; they should not be inferred merely by differentiating an empirical mutual-information plot [14, 39]. Hilberg’s original claim was a reinterpretation of Shannon’s experimental data over a very short range of n ; Dębowski’s relaxed form, which replaces the equality by upper and lower power-law bounds, is the version with meaningful empirical support and the one connected to theorems about long-memory sources [10].

4.2 Cagnetta–Raventós–Ganguli–Wyart (2026)

Cagnetta et al. study a data-limited regime in which model size and optimization are intended not to be bottlenecks [6]; they write P for the token budget, which is D here, and $C(n)$ for the covariance matrix written $\Sigma(n)$ below. Their prediction uses two measured dataset statistics, both indexed by a context length or lag n :

$$H_n - H_\infty \asymp n^{-\gamma}, \quad \|\Sigma(n)\|_{\text{op}} \asymp n^{-\beta}, \quad (14)$$

where $H_n = H(X_{t+1} | X_{t-n+1}^t)$ is the next-token conditional entropy at context length n and $\Sigma(n)$ is the token–token covariance matrix at lag n . The argument has two steps. First, since the empirical covariance at lag n is estimated with noise $O(D^{-1/2})$, a correlation of size $n^{-\beta}$ is statistically resolvable only while $n^{-\beta} \gtrsim D^{-1/2}$, so the largest usable horizon is

$$n^*(D) \asymp D^{1/(2\beta)}. \quad (15)$$

Second, if within-horizon excess errors decay faster than the error caused by truncating the context at n^* , the loss is set by the conditional-entropy gap at that horizon:

$$\mathcal{L}(D) - H_\infty \asymp (n^*(D))^{-\gamma} \asymp D^{-\gamma/(2\beta)}, \quad \alpha_D = \frac{\gamma}{2\beta}. \quad (16)$$

The fast within-horizon learning condition is substantive rather than technical: it is what restricts the prediction to models expressive enough to exploit the horizon they can resolve, and the authors expect it to fail for shallow networks, kernels, or n -gram methods. Their reported estimates are

$$\begin{aligned} \text{TinyStories: } \gamma &= 0.325 \pm 0.003, \quad \beta = 0.88 \pm 0.06 \quad \Rightarrow \quad \alpha_D = 0.185 \pm 0.013, \\ \text{WikiText: } \gamma &= 0.265 \pm 0.016, \quad \beta = 0.94 \pm 0.16 \quad \Rightarrow \quad \alpha_D = 0.141 \pm 0.025, \end{aligned} \quad (17)$$

with the intervals obtained by bootstrapping and standard error propagation, and with no parameters fitted to the learning curves themselves. Empirical exponents from GPT-2- and LLaMA-style transformers fall inside those intervals over the tested ranges, and the loss curves collapse under the rescaling $D \mapsto D/n^{2\beta}$, $\mathcal{L}_n \mapsto n^\gamma \mathcal{L}_n$ implied by the derivation. Their γ estimates are somewhat above the value near 0.23 obtained by compression-based extrapolation [39], on different corpora.

This result is unusually direct because it predicts an exponent from independently measured corpus statistics rather than from a synthetic data model, and because both inputs are falsifiable on any new corpus. Equation (16) also has discriminating content: corpora sharing γ but differing in correlation decay β should have different data exponents. Three limitations bound the claim. It relies on power-law fits over finite ranges, and the WikiText correlation exponent

in particular carries a ± 0.16 uncertainty that dominates its prediction interval. It estimates H_n using trained models as upper bounds, so γ inherits whatever bias those models carry. And the collapse is not uniformly sharp: for LLaMA on TinyStories the visually best collapse occurs near $\beta \simeq 0.65$, outside the independently measured range, which the authors report and set aside as a finite-range or architecture-dependent effect. It also targets the data-limited regime only, and says nothing about parameter or compute scaling.

5 Geometry, regimes, and approximation limits

5.1 The manifold-tiling mechanism

Suppose a smooth target on a compact d -dimensional manifold is approximated by affine pieces on K cells of typical diameter $K^{-1/d}$. If the pointwise Taylor remainder is second order and loss is squared error, then

$$|f - \hat{f}| = O(K^{-2/d}), \quad \mathbb{E}|f - \hat{f}|^2 = O(K^{-4/d}). \quad (18)$$

If a network provides $K \asymp N$ effective cells, Equation (18) suggests $\alpha_N \approx 4/d$ [35].

The last step— $K \asymp N$ —is a hypothesis, not a general neural-network theorem. Deep ReLU networks can have exponentially more linear regions than parameters [25], learned regions need not tile uniformly or align with the data manifold, classification cross-entropy is not globally equivalent to smooth regression, and representation dimension can vary with scale.

Sharma and Kaplan explicitly call the transfer to realistic networks conjectural, and their own language-model evidence supports an inequality rather than the equality $\alpha_N = 4/d$. For GPT-type models with $\alpha_N \approx 0.076$ one has $4/\alpha_N \approx 53$, while intrinsic-dimension estimates from last-token activations exceed 90; the authors therefore conclude only that $d \geq 4/\alpha_N$, and note it “would have been more exciting” to find equality. Their measurements also undercut the premise that a single d exists: the estimated dimension of activations drawn from one contiguous 1024-token passage is about 7, an order of magnitude below the estimate across passages, which indicates a scale-dependent manifold rather than one intrinsic dimension. Claims occasionally attributed to this work that language representations have $d \approx 10$ –30 do not appear in it.

5.2 Four scaling regimes, not a universal two-term envelope

Bahri et al. distinguish variance- and resolution-limited behavior along both model and data axes, producing four regimes [1]:

Varied resource	Limiting mechanism	Behavior in their analysis
data D	finite-sample variance near an infinite-data limit	correction $O(D^{-1})$
width w	finite-width variance near an infinite-width limit	correction $O(w^{-1})$
data D	resolution of a smooth d -manifold	exponent proportional to $1/d$
parameters or features P	resolution of a smooth d -manifold	exponent proportional to $1/d$

The D^{-1} and width results follow from the existence of a well-behaved infinite-data or infinite-width limit plus averaging, and are observed with exponent very close to 1 in their controlled experiments; the paper does not claim every data-limited neural curve has exponent one, and the four regimes are alternatives rather than additive contributions. What is proved in the resolution-limited regime is weaker than what is usually quoted from it. Their Theorems 2 and 3 assume the target, the model, and the loss are Lipschitz and that the model interpolates the training points, and conclude only $L = O(D^{-1/d})$ and $L = O(P^{-1/d})$, where P is a parameter or feature count; the argument is essentially a nearest-neighbour distance bound on

a d -manifold. The stronger $D^{-c/d}$ with $c \geq 2$ —the form that reproduces the $4/d$ relation—is offered as a typical-case estimate under analyticity of the target and the loss, with c set by the leading nonvanishing Taylor order, not as a theorem. Likewise, the smooth “harmonic crossover” between regimes used in scaling-law fitting is an empirical ansatz [5], not a consequence of this taxonomy.

5.3 What lower bounds actually constrain

Continuous nonlinear widths. DeVore, Howard, and Micchelli prove worst-case lower bounds for approximation of smoothness balls when the encoder selecting N real parameters is continuous [11]. For Sobolev-type classes the characteristic norm-error rate is $N^{-s/d}$ under the theorem’s embeddings and parameter ranges. This does not imply that gradient descent on every neural architecture has that rate, in either direction. One must establish that the learned encoder/decoder falls within the relevant continuous-width model, and the result is worst-case over a whole function ball rather than distribution-specific. The continuity hypothesis is doing real work: Yarotsky showed that deep ReLU networks whose weights are allowed to depend discontinuously on the target achieve rates strictly better than the continuous-selection bound, improving the exponent in the weight count W by a factor of two for a fixed modulus of continuity [43]. An approximation lower bound is therefore a statement about a parameter-selection model, not an intrinsic ceiling on networks of a given size.

Parametric redundancy. For regular identifiable k -dimensional parametric families, universal coding and Bayesian redundancy have a leading cumulative term $(k/2) \log n + O(1)$ under suitable compactness and Fisher-information assumptions [29, 30]. Per observation this is $O(k \log n/n)$. Singular and overparameterized neural models need not obey the regular-model constant, so the result cannot be inserted with $k = N$ without further analysis.

Statistical-query limitations. Kearns introduced the statistical-query (SQ) model and showed that parity functions under the uniform distribution are not efficiently learnable by polynomially many inverse-polynomial-tolerance SQs [21]. This is an unconditional limitation, but only on SQ algorithms. Gradient descent on a finite sample is not an SQ algorithm—among other things, it can and does read individual examples at full precision—so the bound transfers only under an explicit noise or precision model. It is therefore not a theorem that all neural networks require $2^{\Omega(d)}$ samples or parameters, and it does not on its own imply a cryptographic hardness result for unrestricted learners. Statements combining SQ and cryptographic lower bounds into a single “neural-network hardness theorem” conflate two distinct models and are not supported by either source.

6 Spectral and solvable models

6.1 Kernel learning curves require target alignment

For a kernel integral operator with eigenpairs (λ_k, ψ_k) and target $f^* = \sum_k c_k \psi_k$, the learning curve depends on the full pairs (λ_k, c_k^2) , as well as label noise and regularization. Bordelon et al. derive approximate typical-case modal learning curves, and Canatar et al. analyze spectral bias, task–model alignment, and possible non-monotonicity [4, 7]. No universal exponent is determined by the eigenvalue decay $\lambda_k \asymp k^{-a}$ alone: two targets with the same kernel and different alignment profiles c_k^2 have different learning curves, which is the content of the task–model alignment analysis. Exponent formulas quoted from this literature must therefore always be read together with the convention for c_k^2 , the noise level, and the regularization; a formula stated in terms of a and b alone, with no such qualification, is not a general consequence of these papers.

What can be said cleanly is a hard-cutoff heuristic, and it is worth stating carefully because the cutoff—not the tail sum—is where the algorithm enters. Suppose the learner resolves the modes up to some index k_* and leaves the rest untouched. The residual error is then the unlearned target power, and with $c_k^2 \asymp k^{-b}$, $b > 1$,

$$\mathcal{E} \approx \sum_{k > k_*} c_k^2 \asymp k_*^{-(b-1)}. \quad (19)$$

For ridgeless kernel regression on n samples the cutoff is set by the effective ridge κ , which solves $n \asymp \sum_k \lambda_k / (\lambda_k + \kappa) \asymp \kappa^{-1/a}$; hence $\kappa \asymp n^{-a}$ and $\lambda_{k_*} \asymp \kappa$ gives $k_* \asymp n$. Samples buy modes roughly one for one, whatever the decay rate a , and Equation (19) gives $\mathcal{E}(n) \asymp n^{-(b-1)}$. Substituting the source/capacity parametrization $\lambda_k = k^{-\alpha}$, $c_k^2 = k^{-(1+2\alpha r)}$ recovers the familiar noiseless kernel rate $n^{-2\alpha r}$ [9].

The word “ridgeless” and the word “noiseless” in that sentence are both load-bearing. Cui et al. show that label noise under optimal regularization gives a different exponent, $n^{-(b-1)/b}$, and that a noisy learning curve crosses over from the noiseless exponent to the noisy one as n grows, so a single power law describes at most one stretch of the curve [9]; the noiseless exponent also saturates once $b - 1 > 2a$. Spigler et al. measure spectrum-driven exponents on empirical data and find they can depart from teacher–student predictions for the same kernel [38]. The reliable general conclusions are spectral ordering—larger-eigenvalue modes are learned first—and dependence on alignment. Equation (19) is a template into which a cutoff must be supplied; it is not itself a learning curve.

6.2 A rigorous linear-regression scaling law

Lin, Wu, Kakade, Bartlett, and Lee study infinite-dimensional linear regression trained by one-pass SGD, with a fixed Gaussian sketch of the covariates playing the role of model size and covariance eigenvalues decaying as $\lambda_k \asymp k^{-a}$, $a > 1$ [22]. In their notation the sketch dimension is M and the sample size is N ; translating to the conventions of Section 1, with N parameters and D samples, their source condition is $c_k^2 \asymp k^{-b}$ in the notation of Section 6.1 and the reducible test error is

$$\Theta\left(N^{-(b-1)} + D^{-(b-1)/a}\right), \quad 1 < b < a + 1, \quad (20)$$

which reduces to $\Theta(N^{-(a-1)} + D^{-(a-1)/a})$ under their isotropic parameter prior, the case $b = a$.

Equation (20) is two instances of the cutoff template (19), and reading it that way shows exactly where the algorithm enters. Sketching to N dimensions truncates at $k_* = N$, giving $N^{-(b-1)}$. One-pass SGD, unlike ridge regression, resolves a mode once the accumulated step size times its eigenvalue is order one, so $D\lambda_{k_*} \gtrsim 1$ and $k_* \asymp D^{1/a}$, giving $D^{-(b-1)/a}$. The same spectrum and the same target thus yield a data exponent of $(b-1)/a$ under one-pass SGD and $(b-1)$ under ridgeless kernel regression: the exponent is a property of the algorithm as much as of the data.

Two further points are worth extracting. The result resolves a puzzle—naively the variance error grows with model size, which would forbid monotone improvement in N , but here the implicit regularization of one-pass SGD makes that term strictly higher order, so it never appears in the rate. And both exponents are set by the pair (a, b) rather than being free: $\alpha_N = b - 1$ and $\alpha_D = (b - 1)/a$, hence $\alpha_N = a\alpha_D$. That is a falsifiable constraint linking two quantities that Chinchilla-style fits estimate independently. Taken at face value, the Hoffmann exponents $\alpha_N = 0.34$ and $\alpha_D = 0.28$ would correspond to $a \approx 1.2$ and $b \approx 1.3$, which lie in the admissible range—so the two are not in conflict, but neither is the agreement evidence for the model until the spectrum of a transformer’s features is measured and compared. Equation (20) is a genuine joint scaling theorem, but for a particular linear model, prior, and training algorithm.

Maloney, Roberts, and Sully provide a complementary solvable random-feature model in which a power-law spectrum produces model-, data-, and compute-scaling regimes [23]. Such

models demonstrate mathematical consistency of scaling mechanisms rather than universality for finite feature-learning networks.

7 Discrete tasks, memorization, and superposition

7.1 Quantization model

Michaud, Liu, Girit, and Tegmark propose that knowledge decomposes into discrete “quanta” learned roughly in decreasing order of frequency [24]. If quantum utilities obey $U_i \asymp i^{-q}$ with $q > 1$ and the first K quanta are mastered, then the remaining loss is the elementary tail

$$\sum_{i>K} U_i \asymp K^{-(q-1)}. \quad (21)$$

If, in addition, $K \asymp N$, this gives $L(N) - E \asymp N^{-(q-1)}$. The tail calculation is exact, and it is Equation (19) again with quantum utilities in place of target power and $k_* \asymp N$ as the supplied cutoff; the decomposition into quanta, the learning order, the independence and uniform capacity cost of quanta, and $K \asymp N$ are all hypotheses. A heavy-tailed utility ranking is plausible for language given the empirical status of Zipf-like frequency distributions [27], but the step from word frequencies to *skill* frequencies is itself an assumption. The paper validates its predictions in toy settings and reports suggestive LLM evidence; it does not prove that training allocates parameters greedily in utility order.

7.2 Hutter’s memorization model

Hutter analyzes a classifier on countably many discrete inputs that memorizes a label after first observing the input [16]. If input probabilities satisfy $\theta_i \propto i^{-(1+\eta)}$ with $\eta > 0$, the expected error over n samples scales as

$$\mathbb{E}E_n \asymp n^{-\eta/(1+\eta)}. \quad (22)$$

The mechanism is again a tail sum: inputs with $n\theta_i \gtrsim 1$ have almost surely been seen, and the error is the residual mass $\sum_{i>i_*} \theta_i$ with $i_* \asymp n^{1/(1+\eta)}$. Exponential input frequencies instead give an n^{-1} rate up to logarithmic factors, and finitely supported distributions eventually give exponential decay—so the power law is a consequence of the heavy tail specifically, not of memorization. The paper is a transparent demonstration that the data-frequency tail controls a memorization learning curve. It does *not* derive deep-learning scaling laws from Levin universal probability, program length, or execution time.

7.3 Feature superposition

Toy models of superposition show how sparse features can share representational dimensions [12]. For an illustrative random dictionary $W \in \mathbb{R}^{d \times M}$ with $M \gg d$ approximately isotropic unit columns and independent centred feature activations of variance $p(1-p)$ (as for a Bernoulli(p) feature that is active with probability p), linear readout of feature i has interference variance

$$\mathbb{E} \left| \sum_{j \neq i} \langle w_i, w_j \rangle x_j \right|^2 = p(1-p) \sum_{j \neq i} \langle w_i, w_j \rangle^2 \approx p(1-p) \frac{M}{d}. \quad (23)$$

The first equality uses independence and zero mean; the second uses $\mathbb{E} \langle w_i, w_j \rangle^2 \approx 1/d$ for random unit columns. Johnson–Lindenstrauss-type constructions show that M vectors can be made nearly orthogonal in dimension $d = O(\log M)$ [19], which is why the interference stays tolerable for sparse features, but that fact alone does not imply the displayed variance: the isotropy and activation assumptions are doing the work. Note also what the expression is and

is not. It says that usable capacity trades off against interference at rate M/d for a *fixed, random* dictionary; it says nothing about which features a trained network chooses to represent, how their importance is distributed, or how the resulting representation error propagates into next-token loss. Superposition is therefore a capacity mechanism, and an account of loss scaling would additionally require a distribution over feature importances—at which point the power law would come from that distribution’s tail, as in Sections 7.1 and 7.2, rather than from superposition itself.

8 Comparative synthesis

Framework	Resource and assumptions	Defensible conclusion	Main limitation
Gaussian rate-distortion	Bit rate; Gaussian source; squared error; power-law covariance	$D(R) \asymp R^{-\rho}$ for $\lambda_k \asymp k^{-(1+\rho)}$ (Thm. 3.1)	Neural parameters are not automatically bits
Metric entropy/minimax	Samples; specified smoothness class, metric, and observation model	Rates such as $n^{-2s/(2s+d)}$ under regular conditions	Class and metric for language modeling are unspecified
Language horizon	Tokens; entropy and covariance decay; fast within-horizon learning	$\alpha_D = \gamma/(2\beta)$ in the horizon-limited regime	Conditional empirical assumptions; data axis only
Manifold tiling	Cells proportional to parameters; smooth target and loss	Piecewise-affine toy model gives $N^{-4/d}$	Transfer to trained networks is conjectural
Kernel spectra	Samples; kernel eigenpairs, target projections, noise, regularization	Modes with larger eigenvalues are learned earlier; ridgeless noiseless cutoff $k_* \asymp n$ gives $n^{-(b-1)}$	Infinite-width or fixed-kernel regime; no eigenvalue-only exponent; noise changes it
Linear SGD	Sketched features and samples; Gaussian prior; power-law covariance	Equation (20): a joint law with $\alpha_N = b - 1$, $\alpha_D = (b - 1)/a$	Specialized linear setting; ties the two exponents together
Discrete quanta	Learned tasks ordered by heavy-tailed utility	A heavy-tailed residual utility sum produces a power law	Quanta and learning order are hypotheses
Superposition	Width; sparse independent features; isotropic dictionary	Interference can scale as M/d in the toy model	Not an end-to-end training or parameter law

Read across the rows, one pattern is more informative than any single entry. In every framework that produces a power law, the exponent traces back to the tail of some ranked quantity—covariance eigenvalues, kernel eigenvalues and target projections, cell counts on a manifold, quantum utilities, input frequencies, or token correlations. The frameworks differ in what they rank and in what they assume about the learner, not in the mathematics that converts a heavy tail into a power law. This is why observing a power law is weak evidence for any particular mechanism, and why the discriminating measurements are of the ranked quantity itself rather than of the loss curve.

9 Open scientific questions

1. **Resource identification.** What operational quantity should replace parameter count in an information-theoretic model: quantized description length, effective rank, Fisher information, algorithmic complexity, or something else?

2. **Feature learning beyond kernels.** Fixed-kernel theory omits representation change. A satisfactory theory must predict how spectra and target alignment evolve during finite-width training.
3. **Dependence and data quality.** Tokens are neither independent nor equally informative. Pruning by example difficulty can improve on power-law data scaling rather than merely shifting its constant [37], while repeating data yields diminishing returns that are themselves predictable and eventually worthless [26]. Both results say that D is the wrong variable; the open question is what replaces it.
4. **Joint non-asymptotic theory.** Useful predictions require finite-range error bars across N, D, C , architecture, optimizer, and context. Pure asymptotics do not explain the breaks, plateaus, and crossovers that are documented empirically [5], and none of the mechanisms surveyed here predicts where a break should occur.
5. **Loss versus downstream behavior.** The relation between smooth log-loss improvement and thresholded task success needs explicit measurement models [32], especially for rare skills and multi-step reasoning where a single token error can fail an entire trajectory.
6. **Test-time computation.** Search, sampling, verification, and external tools create a distinct inference resource that can substitute for pretraining compute on some tasks [36]. Their interaction with pretraining is not captured by $C \simeq 6ND$, and a scaling theory covering both axes does not yet exist.

10 Conclusion

Power laws are mathematically unsurprising when a problem contains scale-free structure: a polynomial spectral tail, a nonparametric smoothness class, long-range correlations, or a heavy-tailed frequency distribution. What remains difficult is identifying which structure controls a particular trained neural network and proving that the learning algorithm exploits it at the predicted rate.

The strongest current results are conditional. Gaussian reverse water-filling, minimax estimation, kernel theory, and linear SGD provide exact or controlled-model statements. The manifold, quantization, and superposition accounts provide useful hypotheses with testable assumptions. The 2026 language-statistics theory narrows the gap for data-limited scaling by connecting an exponent to corpus statistics measured independently of any trained model, and is the clearest template for what progress looks like: name the ranked quantity, measure its tail, predict the exponent, and be wrong in a checkable way. It does not settle parameter or compute scaling.

Two practical consequences follow. First, a fitted exponent is a summary of a range, not a constant of nature; extrapolating it, or reading a fitted floor E as the entropy rate of language, imports assumptions that the fit itself cannot support. Second, adjudicating between mechanisms requires measuring the mechanism’s own primitive—spectra, alignments, intrinsic dimensions, correlation decay, frequency tails—rather than fitting the loss curve more carefully, since all of these mechanisms fit power laws by construction. Accordingly, neural scaling laws should be treated as robust empirical regularities with several candidate explanations—not as inevitable consequences of Shannon theory or as universal limits on intelligence.

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